AMPHORA / FIRST THREE MONTHS LIVE / 14 OCTOBER 2026

Early-stage compute estimates

Target: two concurrent campaigns, supporting operational agents, and 20 newly trained ML nodes per week.
Production: Linux with isolated containerized services and workers. Windows is only the current development workstation.

Budget $6,500/month, or approximately $19,500 for the first three months.
Use one primary cloud, protect campaign capacity, and rent training compute only when jobs are ready. Keep a GPU burst option available rather than committing to a permanent GPU fleet.

What that budget buys

Monthly allocation	Ballpark cost
Control services, two campaign worker pools, retrieval and agent runners	$1,200
Managed PostgreSQL with standby	$750
Storage, backups, networking, monitoring and staging	$850
Elastic CPU training for 20 model releases/week	$1,600
Initial GPU burst allowance	$150
Campaign and operational-agent model/API usage	$750
Contingency / rounding	$1,200
TOTAL	$6,500

Includes infrastructure, training, storage and agent APIs. Engineering/scientific labor, commercial dataset licenses, wet-lab costs and taxes are separate. These are rounded planning allocations; the underlying reference calculation is $6,327/month.

Reasonable spending range

Lean: $2,500-3,500/month with smaller datasets, limited tuning and lower agent activity. **Recommended: $5,500-7,500/month** for larger classical-ML training and protected campaign capacity. **Heavy expansion: $10,000+/month** for broad tuning, large feature matrices or frequent GPU/model calls.

The $1,000 end of the original range is plausible for a small pilot using existing or inexpensive rented hardware. It is too tight for the larger training program and protected live capacity assumed here.

90-DAY INFRASTRUCTURE PLAN

Capacity: CPU first, GPU when the model needs it

The inherited Amphora models mostly use RDKit descriptors/fingerprints with LightGBM, histogram boosting, random forests, ridge and related classical methods. The separate CrystalAtlas code also uses GPU-backed CatBoost/XGBoost/LightGBM and an 8,000+ descriptor feature bank. The training mix therefore matters more than the count of nodes alone.

Capacity lane	Suggested starting size	Why
Control / coordination	8 vCPU, 32 GB RAM	Keep APIs, durable coordination and the UI responsive.
Two campaigns	One 8-vCPU / 32-GB worker allocation per campaign	Protect live inference from training contention.
CPU training	Up to four 16-vCPU / 128-GB workers; burst to eight when needed	Parallelize trials/folds and scale down after jobs finish.
GPU training	Rent one compatible 24-48-GB GPU as needed; use 80 GB for larger jobs	Supports GPU-coded tree models and later learned molecular models.
Storage	Start with ~3 TB artifacts plus backup capacity and job-local scratch	Reuse dataset/features; retain accepted models and campaign evidence.

How 20 models/week turns into the training estimate

The base assumes **10 trial configurations x 5 validation folds + one final fit** per model: 51 fits. At 15 minutes per fit plus 1.5 hours of preprocessing, each release uses 14.25 worker-hours. Twenty per week, plus 20% job overhead, is about **342 worker-hours/week or 1,482/month**. At roughly $1/worker-hour, training is about $1,600/month.

Four training slots provide enough weekly headroom for this example while remaining elastic. If training must finish inside a two-day weekly window, substantially more burst capacity is needed. The same workload spread through the week is cheaper to provision.

The existing models give us a useful lower bound

A historical BigSolDB run used 94,532 rows and 433 features and recorded about 4-5 minutes for its bounded validation pipeline. Other reviewed property datasets were much smaller. The budget above allows for growth toward roughly one million rows and more tuning; it does not assume every inherited model needs hours on a GPU.

Memory deserves attention: one million rows x 8,000 float64 features is 64 GB before copies and model overhead. Cache features and avoid rebuilding them for every trial. Before launch, benchmark three representative training jobs and two campaigns together. Two-campaign scheduling remains a software delivery item; the production OS target is Linux.

90-DAY INFRASTRUCTURE PLAN

Provider options: practical tradeoffs

Option	Cost anchors / fit	Pros	Cons
AWS	16 vCPU / 128 GB Linux: $1.06/h. L4 GPU VM: ~$0.80-0.98/h. Prices	Broad managed database, storage and batch options. Good single-provider starting point.	Networking, monitoring and data-transfer charges need attention.
Azure	16 vCPU / 128 GB Linux: $0.90/h. Selected A100 VM: $3.67/h. Prices	Competitive CPU prices; attractive with existing Microsoft agreements or expertise.	GPU quotas/availability and complete managed-service pricing must be checked.
GCP	16 vCPU / 128 GB: $1.05/h. L4 VM: ~$0.71/h. Prices / GPU	Competitive GPU examples; good fit where data and ML pipelines already use GCP.	Data location and transfers can outweigh a small compute-price advantage.
ML-focused clouds	Runpod, Lambda, Hyperstack: selected 48-GB GPUs ~$0.50-1.09/h; A100 examples ~$1.35-1.99/h. Runpod / Lambda / Hyperstack	Economical GPU bursts and simple ML environments.	Stock, region, memory, persistent storage and data access differ. Use an appropriate production tier.
Vendor-hosted hardware	CPU servers from OVHcloud or Hetzner; GPU servers also available. Full platform planning range ~$3,500-5,500/month; quote required.	Predictable fees, local NVMe and strong CPU/RAM value at sustained utilization.	Pay when idle; resizing and replacement are slower. More server operations fall to us.
Hardware rented on-site	Obtain a 90-day equipment quote; example vendor.	Local data and direct hardware control.	Add delivery, power, cooling, UPS, connectivity, support and equipment swaps.

Cloud comparison: holding other allowances constant, the reference shape is roughly $6.3k/month on AWS, $5.8k on Azure and $6.3k on GCP. These are indicative comparisons; actual job speed, database configuration and commercial terms determine the final bill.

GPU price comparison: the devices and included CPU/RAM differ. Compare the cost to finish the actual job, not just dollars per GPU-hour. Public prices were checked 14 October 2026, using Linux/on-demand examples and US regions where available. Links above open the primary pricing sources.

Keep the core platform in one place. A specialist GPU vendor can supply occasional workers without moving the databases, campaign history or artifact store. Start with at most one such supplier.

90-DAY INFRASTRUCTURE PLAN

Recommendation and the first 90 days

Choose the primary platform once

Default recommendation: AWS for the core Linux platform, with elastic CPU training. Azure is equally viable and slightly cheaper in the selected CPU examples; choose it if commercial terms or existing expertise favor it. Choose GCP if the team already works there. A few hundred dollars of monthly VM savings should not drive a mid-quarter infrastructure migration.

Rent GPUs, then reserve only what we actually use

GPU training should expand with the selected model mix. Twenty models/week averaging 20 GPU-hours each would require about **2,200 GPU-hours/month** including overhead. That adds roughly **$4k-8k/month** at the compared GPU/VM rates, before the core platform. Occasional tree-model training will be much less.

Dedicated hardware becomes attractive when utilization is sustained. Illustratively, two CPU training servers costing $900/month combined, plus $250 setup/operations allowance, break even against ~$1/hour cloud workers at roughly 1,100 worker-hours/month. Get a quote and benchmark the actual hardware; cloud vCPUs and physical CPU cores are not interchangeable.

Three things that can blow the budget

Driver	Business implication
Unbounded model tuning	Doubling trial counts nearly doubles training work. Give every model a bounded training budget.
Agents reasoning continuously	Five roles calling a model every minute can cost ~$8k/month at the example token mix. Event-driven calls fit the $750 reserve far better. Mini pricing / Escalation pricing
Repeated dataset transfers	Moving 250 GB out to a remote trainer for every monthly release can cost about $2k/month. Cache dataset revisions and train near the data.

A simple rollout

Before launch: benchmark the representative model mix, select the provider/region, confirm capacity and price the full configuration. **Month 1:** use on-demand workers and record real hours, memory, GPU demand and agent spend. **Month 2:** right-size, reuse features and clean up temporary jobs/containers. **Month 3:** consider short reservations or hardware rental if measured usage supports them.

Business planning number: $6,500/month, with a $10,000 ceiling for agreed bursts.
This is a sensible starting envelope for two campaigns and 20 weekly classical-ML releases, while leaving room to grow the GPU lane. Validate the model mix before committing to a fixed fleet.

Assumptions: ~87 model releases/month; bounded tuning; cloud workers scale down between jobs; production Linux containers; storage and agent APIs included. Budget source: v1 model code/receipts plus the linked current vendor prices. Labor, scientific data licenses and tax remain separate.